Lab: Connecting MySQL Workbench to a MySQL Docker Container To begin with the Lab

1. Summary

In this lab, we will learn how to connect MySQL Workbench running on Windows to a MySQL database running inside a Docker container.

In the previous lab, we created a MySQL container using the official MySQL Docker image and configured the root password using the MYSQL_ROOT_PASSWORD environment variable.

We mapped port 3306 of the container to port 3306 on our local machine. This allows external tools, such as MySQL Workbench, to connect to the database server.

MySQL Workbench provides a graphical user interface (GUI) that allows us to manage databases, create tables, execute SQL queries, and view database information.

Understanding MySQL Workbench

MySQL Workbench is a free graphical database management tool provided by MySQL.

Instead of executing every database operation through the Ubuntu terminal, we can use MySQL Workbench to interact with the database visually.

It allows us to:

- Connect to a MySQL database server.
- Create and delete databases.
- Create and modify tables.
- Insert, update, and delete records.
- Execute SQL queries.
- View database structures and information.

Example:

Suppose we have a MySQL database running inside a Docker container.

We can connect to it using MySQL Workbench installed on Windows.

The connection uses:

- Hostname: 127.0.0.1
- Port: 3306

- Username: root
- Password: The password configured when creating the container.

This works because Docker maps the container's port 3306 to the local machine's port 3306.

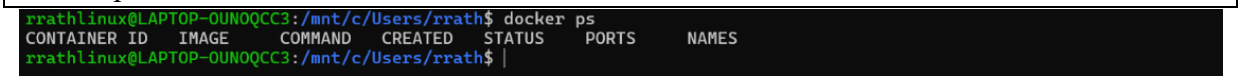
Lab Steps

Step 1: Start the MySQL Container

• Open your Ubuntu WSL terminal.

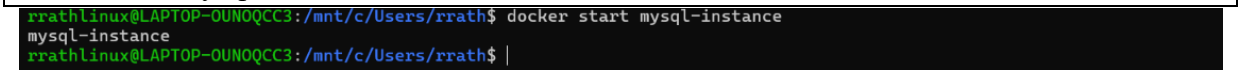
• Check whether the MySQL container is running:

```
docker ps
```



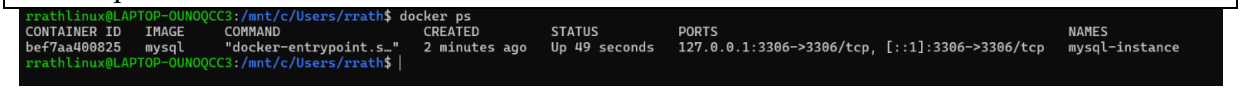
• If the container is stopped, start it:

```
docker start mysql-instance
```



• Confirm that port 3306 is mapped:

```
docker ps
```



Expected port mapping:

0.0.0.0:3306->3306/tcp

Step 2: Download MySQL Workbench

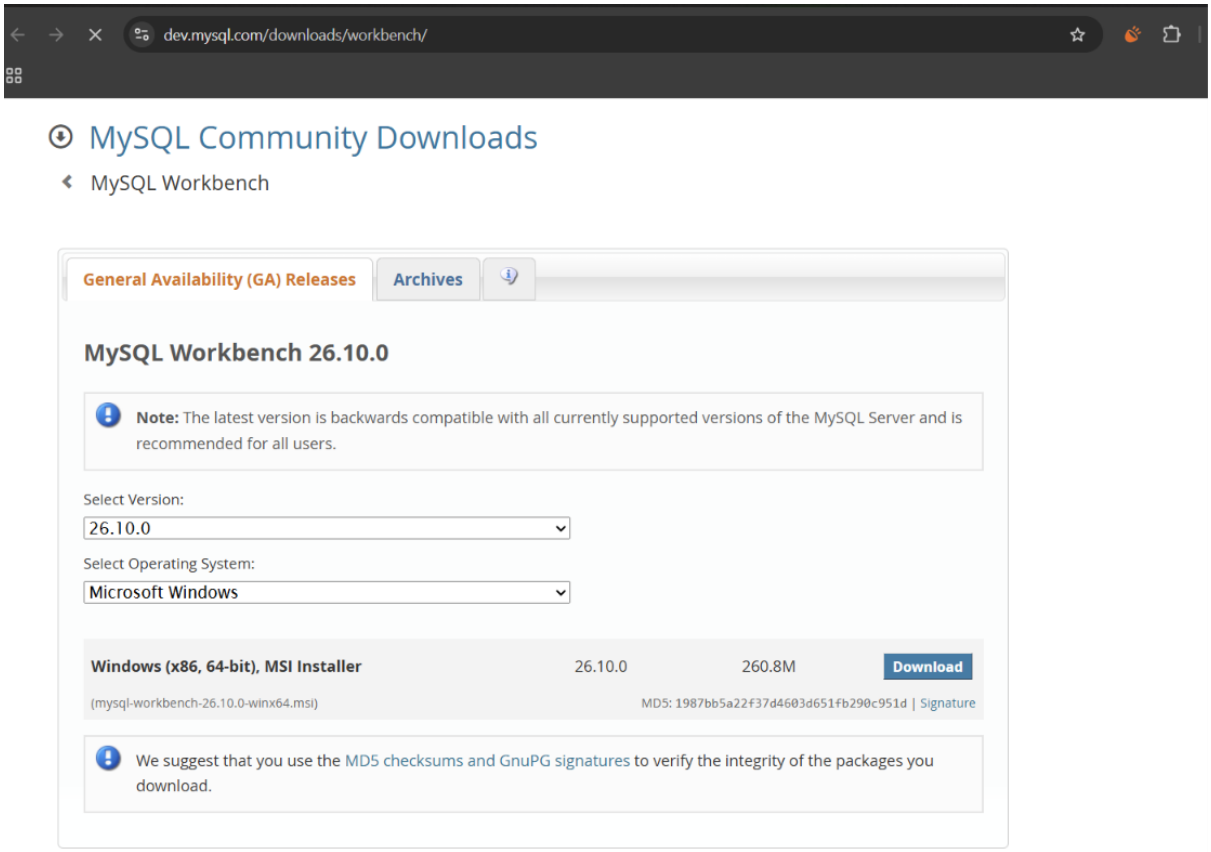
• Open the official MySQL website.

• Navigate to the MySQL Workbench download page.

• Download the Windows installer.

• Run the installer and follow the installation instructions.

Official download: MySQL Workbench Downloads

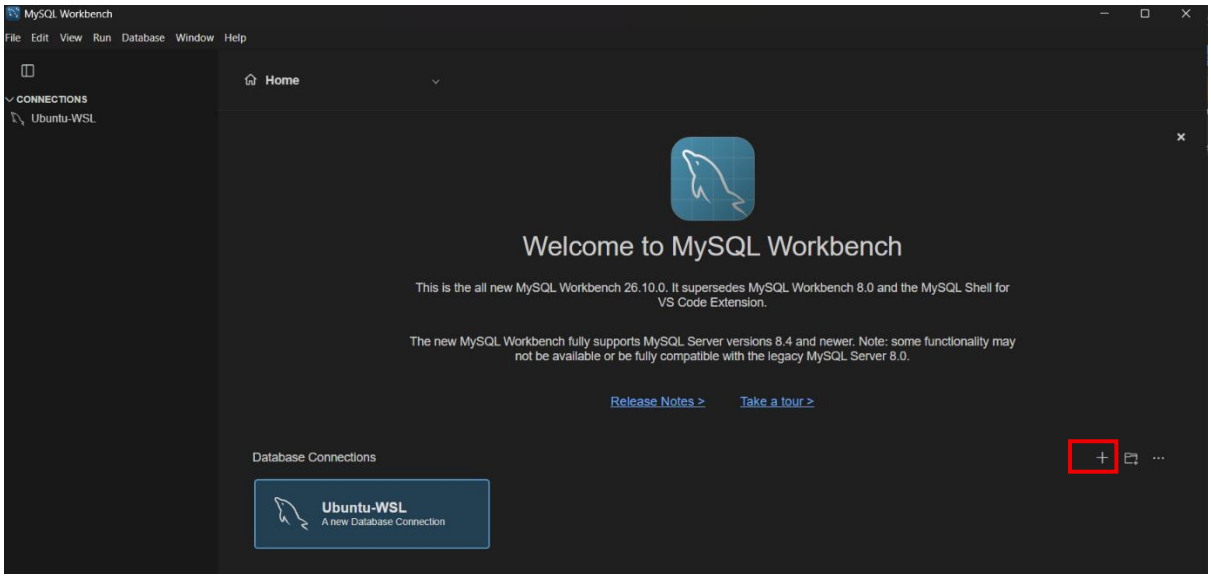


Step 3: Open MySQL Workbench

• Open MySQL Workbench on Windows.

• Locate the MySQL Connections section.

• Click the + icon to create a new connection.

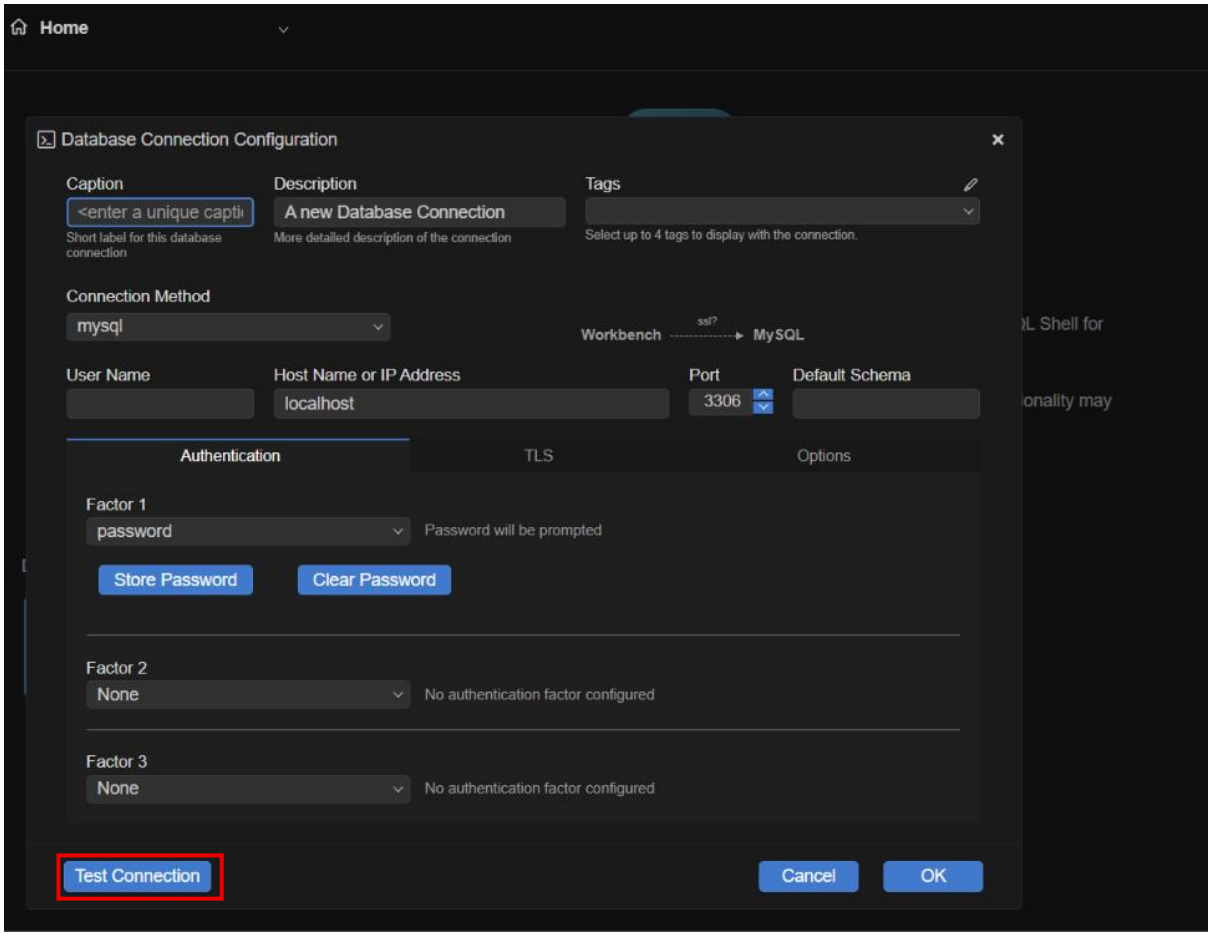


Step 4: Configure a New MySQL Connection

Enter the following connection details:

Fill the detail .

Click on the Test connection.



Setting	**Value**
Connection Name	MySQL-Docker
Connection Method	Standard TCP/IP
Hostname	127.0.0.1
Port	3306
Username	root
Password	Your configured MySQL root password

• Click **Test Connection** to verify the connection.

• If successful, click **OK** to save the connection.

Step 5: Connect to the MySQL Database

• Click the newly created connection.

• Enter your MySQL root password when prompted.

• Click **OK**.

You should now see the MySQL Workbench interface connected to your MySQL database server running inside the Docker container.

Step 6: Execute SQL Queries

Once connected, you can execute SQL commands through the Workbench query editor.

